

Entanglement and Bell States

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Example: Bell states

Can the following state be represented as a product state?

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

No!

This is an entangled state. It cannot be separated into a product of two states.

This state can exhibit “nonlocal” correlations.

Entanglement

Unentangled

Alice

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Bob

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)(|0\rangle + |1\rangle)$$

$$q_{AB} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \begin{matrix} \text{-- "00"} \\ \text{-- "01"} \\ \text{-- "10"} \\ \text{-- "11"} \end{matrix}$$

$$\Pr(00) = \Pr(01) = \Pr(10) = \Pr(11) = \frac{1}{4}.$$

Entangled

Alice

Bob

cannot write separately

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

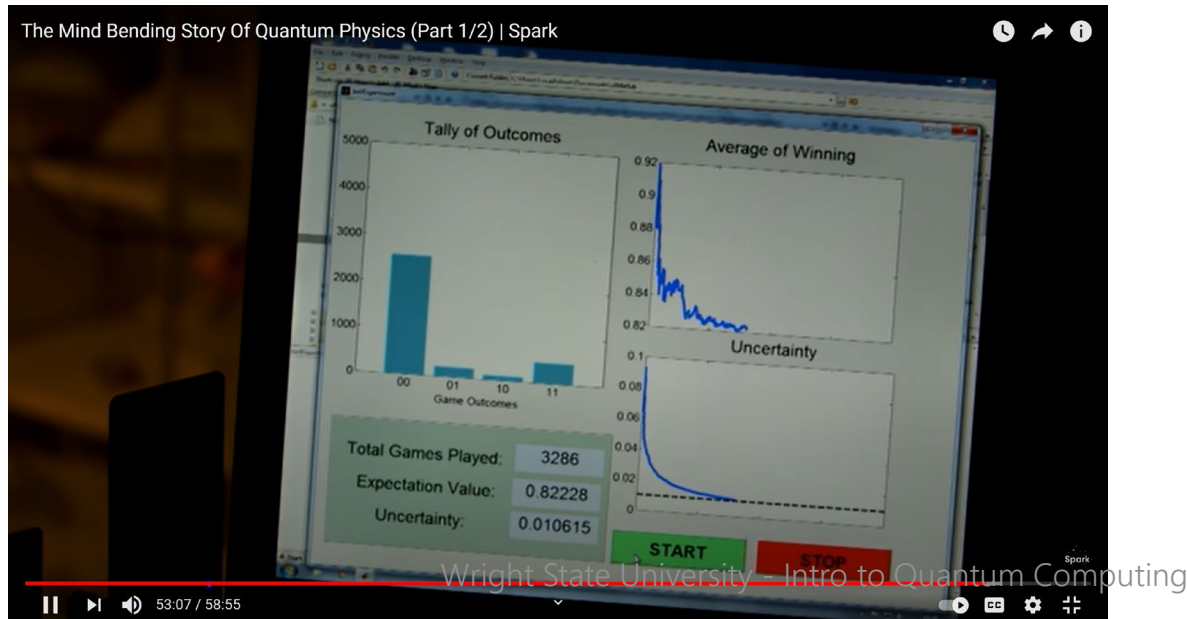
$$q_{AB} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \begin{matrix} \text{-- "00"} \\ \text{-- "01"} \\ \text{-- "10"} \\ \text{-- "11"} \end{matrix}$$

$$\Pr(00) = \Pr(11) = \frac{1}{2}.$$

$$\Pr(01) = \Pr(10) = 0.$$

What's a Bell State?

- Watch more of the Al Khalili Documentary
- Einstein's Nightmare
- 33:33 to 54:20
- <https://www.youtube.com/watch?v=ISdBaf-ysl0&t=2717s>





Example: orthogonal measurement

Consider the state $|q\rangle = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$. Measure $|q\rangle$ in the $|+\rangle$ basis.

Recall $|+\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Thus $|-\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

$|+\rangle, |-\rangle$ make an orthogonal basis.

Writing $|q\rangle$ in this basis: $|q\rangle = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = c_0|0\rangle + c_1|1\rangle = c_0 \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) + c_1 \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$
$$= \frac{1}{\sqrt{2}}((c_0 + c_1)|+\rangle + (c_0 - c_1)|-\rangle)$$

Prob measuring $|+\rangle$: $\left| \frac{1}{\sqrt{2}} (c_0 + c_1) \right|^2 = \frac{|c_0 + c_1|^2}{2}$

Prob measuring $|-\rangle$: $\left| \frac{1}{\sqrt{2}} (c_0 - c_1) \right|^2 = \frac{|c_0 - c_1|^2}{2}$



Orthogonal measurement

Consider orthogonal basis vectors $|b_0\rangle$ and $|b_1\rangle$.

$$|q\rangle = c_0|b_0\rangle + c_1|b_1\rangle$$

Then we have

- Outcome b_0 with probability $|c_0|^2$
- Outcome b_1 with probability $|c_1|^2$

Lecture Summary so far

- ▶ **Superposition**: is *key for quantum advantage*.
 - ▶ The qubit state is $\alpha|0\rangle + \beta|1\rangle$ (a combination of *both* $|0\rangle$ and $|1\rangle$ with complex numbers α and β).
- ▶ **Entanglement**: is *key for quantum advantage*.
 - ▶ Separated qubits can exist in states that are correlated in a way that cannot be replicated classically.
- ▶ **Quantum Measurement**: is quantum readout. It *restricts quantum advantage* as we cannot access the c_i .
 - ▶ Measurement collapses the state to *either* $|0\rangle$ or $|1\rangle$. Crucially, qubits must be protected from unintended measurement.
- ▶ **Quantum Gates**: are used to process quantum information. These will be the subject of our lecture next week.
- ▶ Next videos for this week is about Amplitude and Probability, and then some Quantum Tools, particularly getting an account to use IBM Quantum Computers.